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Definite Integration

NCERT frankly ask Questions (Board Level)

Question

Evaluate
$$\int_0^1 (3x^2 + 4x + 2)\,dx$$

Concept Used

Fundamental Theorem of Calculus

Solution

$$\int (3x^2 + 4x + 2)\,dx = x^3 + 2x^2 + 2x$$

Applying limits,

$$\left[x^3 + 2x^2 + 2x\right]_0^1 = 1 + 2 + 2 = 5$$

Final Answer

$$\boxed{5}$$

Question

Evaluate
$$\int_0^{\pi/2} (\sin x + \cos x)\,dx$$

Concept Used

Basic Trigonometric Integration

Solution

$$\int (\sin x + \cos x)\, dx = -\cos x + \sin x$$

Hence,

$$[-\cos x + \sin x]_0^{\pi/2} = (0 + 1) - (-1 + 0) = 2$$

Final Answer

$$\boxed{2}$$

Question

Evaluate

$$\int_0^1 (2x + 1)^4\, dx$$

Concept Used

Substitution Method

Solution

Let

$$u = 2x + 1, \qquad du = 2dx$$

$$dx = \frac{du}{2}$$

Limits become

$$x = 0 \Rightarrow u = 1, \qquad x = 1 \Rightarrow u = 3$$

Hence,

$$I = \frac{1}{2} \int_1^3 u^4 du$$

$$= \frac{1}{2} \left[\frac{u^5}{5} \right]_1^3$$

$$= \frac{1}{10}(243 - 1) = \frac{121}{5}$$

Final Answer

$$\boxed{\frac{121}{5}}$$

Question

Using the property of definite integrals, evaluate

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx$$

Concept Used

Property

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

Solution

Let

$$I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

Using the property,

$$I = \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$$

Adding,

$$2I = \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

Put

$$u = \cos x, \qquad du = -\sin x dx$$

Therefore,

$$2I = \pi \int_{-1}^{1} \frac{du}{1 + u^2}$$

$$= \pi \left[\tan^{-1} u\right]_{-1}^{1} = \frac{\pi^2}{2}$$

Hence,

$$I = \frac{\pi^2}{4}$$

Final Answer

$$\boxed{\frac{\pi^2}{4}}$$

Question

Evaluate

$$\int_0^{\pi/2} \sin^2 x \, dx$$

Concept Used

Identity

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

Solution

$$I = \frac{1}{2} \int_0^{\pi/2} (1 - \cos 2x) dx$$

$$= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi/2}$$

$$= \frac{1}{2} \left(\frac{\pi}{2} \right) = \frac{\pi}{4}$$

Final Answer

$$\boxed{\frac{\pi}{4}}$$